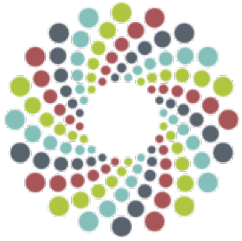
~~Increasing Openness and Connections throughout the Scientific Workflow~~

The Importance of APIs in RDM

(and other assorted acronyms)



figshare

The Dataverse Project

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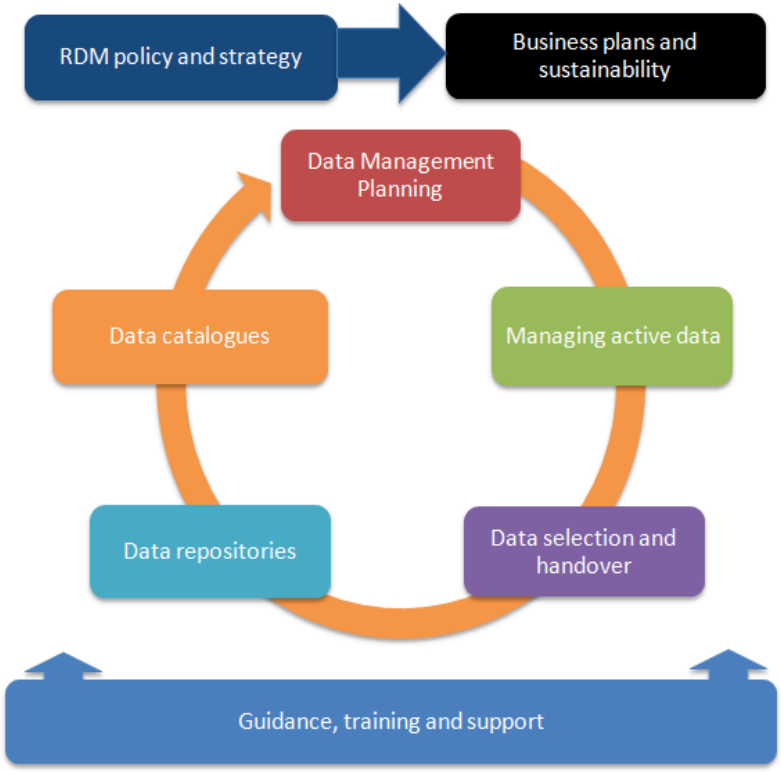
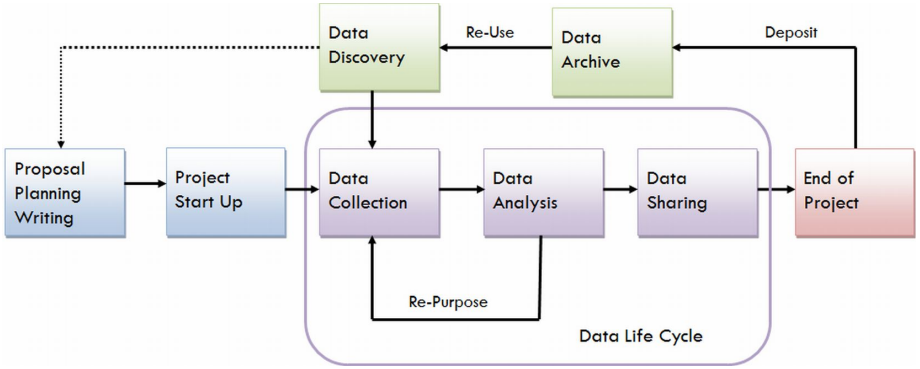
Recommended paths for RDM





Van den Eynden, V., Corti, L., Woollard, M. & Bishop, L. (2009). *Managing and Sharing Data: A Best Practice Guide for Researchers*. Retrieved 02/06/2010, from http://www.data-archive.ac.uk/media/2894/managingsharing.pdf

Jones, S., Pryor, G. & Whyte, A. (2013). 'How to Develop Research Data Management Services - a guide for HEIs'. DCC How-to Guides. Edinburgh: Digital Curation Centre. Available online: http://www.dcc.ac.uk/resources/how-guides - See more at: http://www.dcc.ac.uk/resources/how-guides/how-develop-rdm-services#sthash.xYof7KTN.dpuf

Strasser, Carly; Cook, Robert; Michener, William; & Budden, Amber. (2012). Primer on Data Management: What you always wanted to know. UC Office of the President: California Digital Library. Retrieved from: http://escholarship.org/uc/item/7tf5q7n3

101 Innovations in Scholarly Communication

The Changing Research Workflow

Jeroen Bosman @jeroenbosman
Utrecht University Library

Bianca Kramer @MsPhelps
Utrecht University Library

Science is in transition. This poster gives an impression of the exploratory phase of a project aiming to chart innovation in scholarly information and communication flows from evolutionary and network perspectives.

We intend to address the questions of what drives innovation and how these innovations change research workflows and may contribute to more **open, efficient** and **good** science.

101 Innovative tools and sites in 6 research workflow phases (< 2000 - 2015)

Most important developments in 6 research workflow phases

	Discovery	Analysis	Writing	Publication	Outreach	Assessment
Trends	social discovery tools	data-driven & crowdsourced science	collaborative online writing	Open Access & data publication	scholarly social media	article-level (alt)metrics
Expectations	growing importance of data discovery	more online analysis tools	more integration with publication & assessment tools	more use of "publish first, judge later"	use of altmetrics for monitoring outreach	more open and post-publication peer review
Uncertainties	support for full-text search and text mining	willingness to share in analysis phase	acceptance of collaborative online writing	effect of journal/publisher status	requirements of Funders & institutions	who pays for costs qualitative assessment?
Opportunities	discovery based on aggregated OA full text	open laboratory	semantic tagging while writing/citing	reader-side paper formatting	using repositories for institutional visibility	using author-, publication- and affiliation-IDs
Challenges	real semantic search (concepts & relations)	reproducibility	safety/privacy of online writing	globalization of publishing/access standards	making outreach a two-way discussion	quality of measuring tools

Most important long-term development	multidisciplinary + citation-enhanced databases	collaboration + data-driven	online writing platforms	Open Access	more & better connected researcher profiles	importance of societal relevance + non-publication contributions
Potentially most disruptive development	semantic/concept search + contextualized/social recommendations	open science	collaborative writing + integration with publishing	circumventing traditional publishers	public access to research findings, also for agenda setting	moving away from simple quantitative indicators

Typical workflow examples

Open Science

See more: http://innoscholcomm.silk.co/page/Workflows
https://101innovations.wordpress.com/

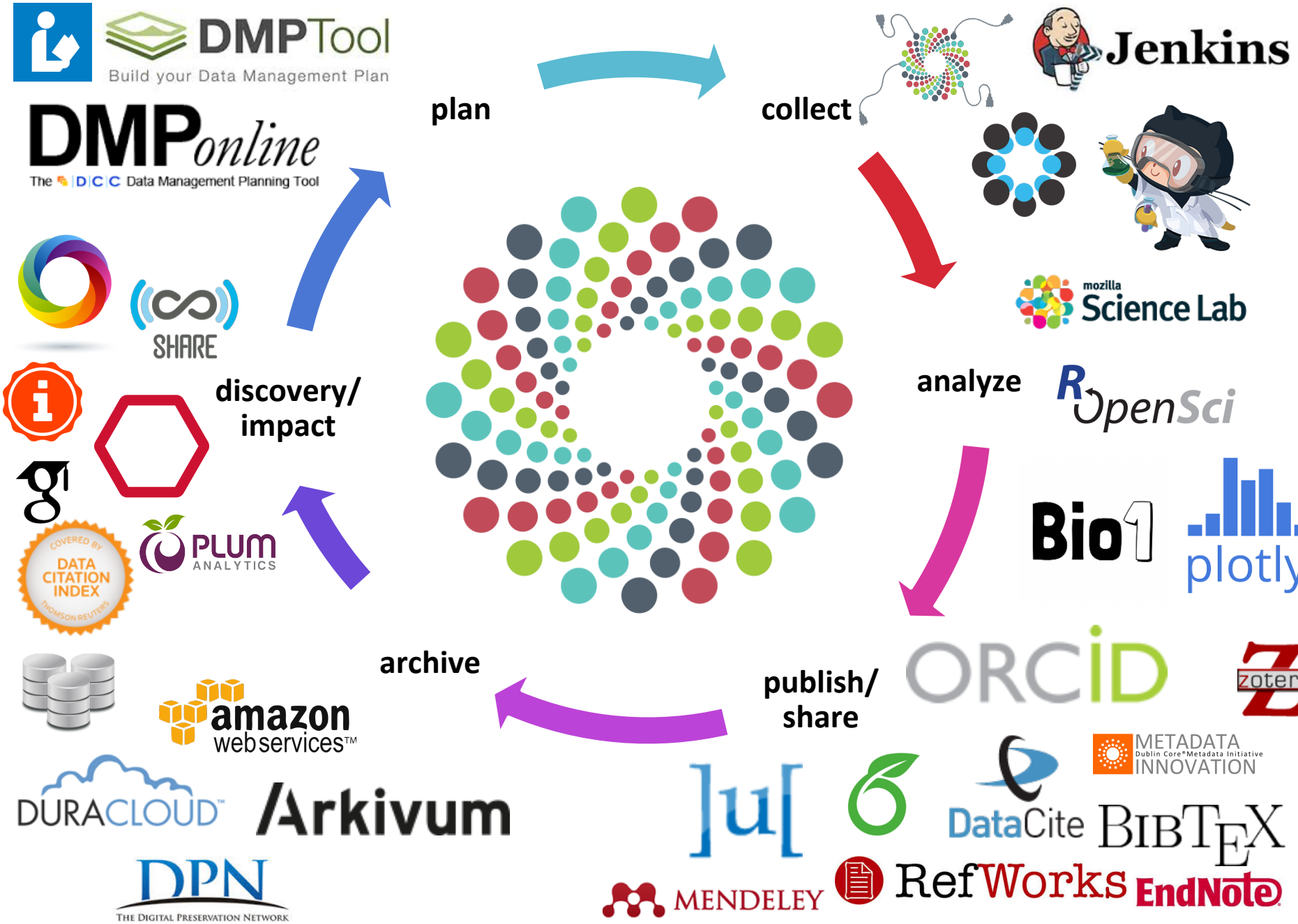
Kramer, Bianca; Bosman, Jeroen (2015): 101 Innovations in Scholarly Communication - the Changing Research Workflow. figshare.
http://dx.doi.org/10.6084/m9.figshare.1286826
Retrieved 16:29, Oct 13, 2015 (GMT)

figshare's API

The figshare API allows users to push data to figshare, or pull data off of the platform

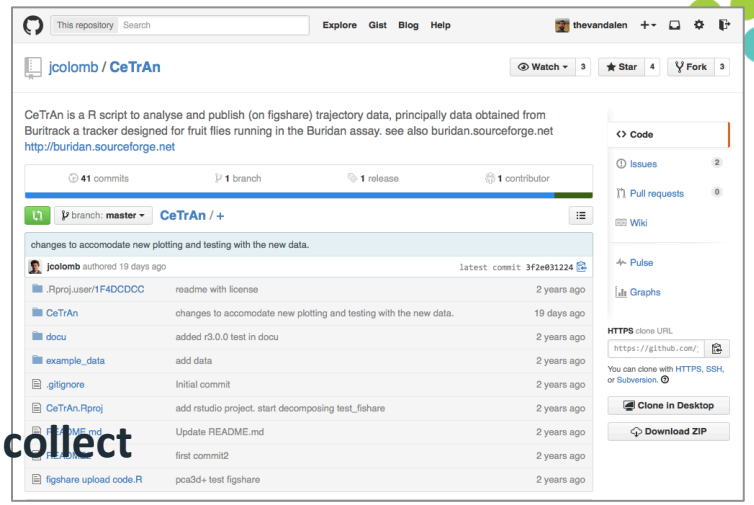
This allows you to build applications on top of academic research

figshare integrations in action



An R open science example…

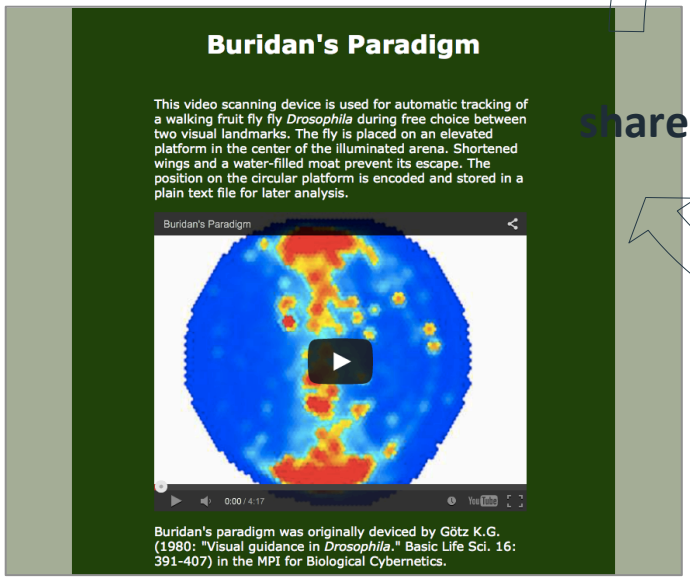
	id	group	genotype	treatment	machine	other	date	timeofday	median_spe	distance_tra	turning_angl	meander	activitytime	act_bouts_S	pause_durat	numb_pause	activitytime_
1	100_CSJC	CS_JC_2ndcr	CS_JC	none	A	2ndcross	9/20/13	##########	17.6122386	510.940185	7.08158437	0.39205073	143.5	3.14999998	1.5	19	138.9
2	101_CSBJC	CS_JC_2ndcr	CS_JC	none	B	2ndcross	9/20/13	##########	13.3093301	231.145092	11.8844266	0.7789247	135.1	2.70000005	1.5	31	135.7
3	102_CSJC	CS_JC_2ndcr	CS_JC	none	C	2ndcross	9/20/13	##########	15.9188682	368.327127	9.29123379	0.55608839	159.3	3.79999995	0.9000001	13	159.2
4	103_CSJC	CS_JC_2ndcr	CS_JC	none	D	2ndcross	9/20/13	##########	16.5038787	717.444774	7.54849732	0.45214688	273.600001	11.3000002	1.04999995	14	271
5	104_CSTP	CS_TP_2ndcr	CS_TP	none	A	2ndcross	9/20/13	##########	15.5501774	777.795393	6.55529888	0.40337478	291.1	6.79999995	0.50000012	8	293.8
6	105_CSTP	CS_TP_2ndcr	CS_TP	none	B	2ndcross	9/20/13	##########	16.1131212	455.115117	7.40149444	0.43970117	150.5	2.89999998	3	17	146.2
7	106_CSTP	CS_TP_2ndcr	CS_TP	none	C	2ndcross	9/20/13	##########	17.2771567	794.539732	6.59162962	0.36481201	272.3	16.35	1.20000005	9	270.6
8	107_CSTP	CS_TP_2ndcr	CS_TP	none	D	2ndcross	9/20/13	##########	16.1620351	812.967926	7.58395534	0.44949885	294	39.3999999	1.04999995	4	294
9	108_CSBVS	CS_BS_2ndcr	CS_BS	none	A	2ndcross	9/20/13	##########	14.8886868	518.828231	6.35658234	0.38641765	229.199999	7.54999995	0.60000014	25	230.8
10	109_CSBvs	CS_BS_2ndcr	CS_BS	none	B	2ndcross	9/20/13	##########	14.2972058	400.366699	5.71643304	0.38361199	175.7	5.95000005	1.64999998	16	171.4
11	110_CSBVS	CS_BS_2ndcr	CS_BS	none	C	2ndcross	9/20/13	##########	16.2253314	657.356598	6.6266186	0.36946413	256.6	8.20000005	0.75	18	258.499999
12	111_CSBVS	CS_BS_2ndcr	CS_BS	none	D	2ndcross	9/20/13	##########	14.042916	639.790077	6.88606922	0.45529103	281.5	26.4499999	1.89999986	9	280.7
13	112_CSJC	CS_HS_2ndcr	CS_HS	none	A	2ndcross	9/20/13	##########	15.0008946	638.145575	5.94732506	0.37019237	256.000001	6.4000001	2.0999999	19	249
14	113_CSHS	CS_HS_2ndcr	CS_HS	none	B	2ndcross	9/20/13	##########	17.5043204	703.351364	5.35418829	0.28268952	236.699999	9.20000005	2.60000002	18	234.5
15	114_CSHS	CS_HS_2ndcr	CS_HS	none	C	2ndcross	9/20/13	##########	10.8642468	174.942796	9.3507279	0.78705473	160.2	17.9000001	0.9000001	11	165.2
16	115_CSHS	CS_HS_2ndcr	CS_HS	none	D	2ndcross	9/20/13	##########	15.0400845	333.16162	5.9165218	0.36168682	142.5	5	4.4000001	21	137.4
17	116_CSTZ	CS_TZ_2ndcr	CS_TZ	none	A	2ndcross	9/20/13	##########	16.4458536	559.448644	6.64151914	0.37078207	198	11.6999999	0.74999988	10	197
18	117_CSTZ	CS_TZ_2ndcr	CS_TZ	none	B	2ndcross	9/20/13	##########	12.8965403	60.2270539	11.6131909	0.8283955	35.4000001	1.29999995	4	11	35.5000002
19	118_CSTZ	CS_TZ_2ndcr	CS_TZ	none	C	2ndcross	9/20/13	##########	21.6350837	832.369509	5.74553669	0.24687918	268.4	3.89999998	0.70000005	29	270
20	119_CSTZ	CS_TZ_2ndcr	CS_TZ	none	D	2ndcross	9/20/13	##########	14.7033046	523.296729	7.74348163	0.49171641	231.2	6.20000005	1.1500001	22	226.9
21	120_CSJC	CS_JC_2ndcr	CS_JC	none	A	2ndcross	9/20/13	##########	16.4713165	641.470823	7.42486007	0.42853449	246.7	6.20000005	0.79999995	19	241.899999
22	121_CSBJC	CS_JC_2ndcr	CS_JC	none	B	2ndcross	9/20/13	##########	18.0458379	685.768452	7.90810691	0.40873037	255.7	3.64999998	0.60000014	29	247.700001
23	122_CSJC	CS_JC_2ndcr	CS_JC	none	D	2ndcross	9/20/13	##########	15.820831	500.481894	7.42417077	0.45075403	220.9	2.60000002	1.29999995	40	208.1
24	123_CSBTP	CS_TP_2ndcr	CS_TP	none	A	2ndcross	9/20/13	##########	15.2803956	491.485263	8.44485159	0.52401448	204.9	5.24999988	1.44999993	28	193.3
25	124_CSTP	CS_TP_2ndcr	CS_TP	none	B	2ndcross	9/20/13	##########	16.9964861	602.66544	8.11011778	0.45299505	225.299999	11.2	1.4000001	15	222.1
26	125_CSTP	CS_TP_2ndcr	CS_TP	none	C	2ndcross	9/20/13	##########	15.9714384	772.39629	6.96620733	0.42136408	273.3	21.6	1.70000005	7	270.8
27	126_CSBVS	CS_BS_2ndcr	CS_BS	none	A	2ndcross	9/20/13	##########	12.6565204	450.47614	7.6914075	0.5550748	255.599999	3.20000005	1.30000007	24	251
28	127_CSBVS	CS_BS_2ndcr	CS_BS	none	B	2ndcross	9/20/13	##########	17.3626962	710.185765	6.25425327	0.33375696	241.3	16.7	2.75	10	238.5
29	128_CSTZ	CS_TZ_2ndcr	CS_TZ	none	A	2ndcross	9/20/13	##########	16.4501325	283.521069	7.37160861	0.42199624	101.7	45.5	46.3	3	101.6
30	129_CSTZ	CS_TZ_2ndcr	CS_TZ	none	B	2ndcross	9/20/13	##########	15.14244	550.273449	8.13010235	0.49868875	214.6	40.5	7.29999995	5	215.9
31	130_CSTZ	CS_TZ_2ndcr	CS_TZ	none	C	2ndcross	9/20/13	##########	16.4412787	718.485579	7.29914751	0.40790347	274.6	7	0.79999995	13	276.2
32	131_CSTZ	CS_TZ_2ndcr	CS_TZ	none	D	2ndcross	9/20/13	##########	16.4092038	489.699153	6.71172445	0.38769446	180.9	19.9499999	4.94999993	8	178.9
33	132_CSJC	CS_JC_2ndcr	CS_JC	none	A	2ndcross	9/26/13	##########	9.42844536	2.6851132	26.3236799	2.30794334	1.5999999	1.5999999	148.65	2	2.29999995
34	133_CSJC	CS_JC_2ndcr	CS_JC	none	B	2ndcross	9/26/13	##########	18.7399411	331.782623	7.33729051	0.37069148	121.5	4.5999999	3.29999995	17	115.8
35	134_CSJC	CS_JC_2ndcr	CS_JC	none	C	2ndcross	9/26/13	##########	17.2576718	135.484593	6.96032396	0.36530732	57.0999999	1.5999999	1.5000001	25	50.8000019
36	135_CSJC	CS_JC_2ndcr	CS_JC	none	D	2ndcross	9/26/13	##########	17.1818269	353.970673	8.388457	0.45681235	146.2	26.5999999	29.5999995	32	137.999999
37	136_CSTP	CS_TP_2ndcr	CS_TP	none	A	2ndcross	9/26/13	##########	13.2731757	424.555242	8.52774683	0.60971847	233.3	8.5999999	0.79999995	21	236.1
38	137_CSTP	CS_TP_2ndcr	CS_TP	none	B	2ndcross	9/26/13	##########	16.9829146	622.850992	8.59558348	0.48577073	222.4	11.8500001	1.89999998	14	217.099999
39	138_CSTP	CS_TP_2ndcr	CS_TP	none	C	2ndcross	9/26/13	##########	14.9235309	392.965722	7.72771836	0.47208038	183.3	4.45000005	2.0999999	27	172.1



plan

collect

share



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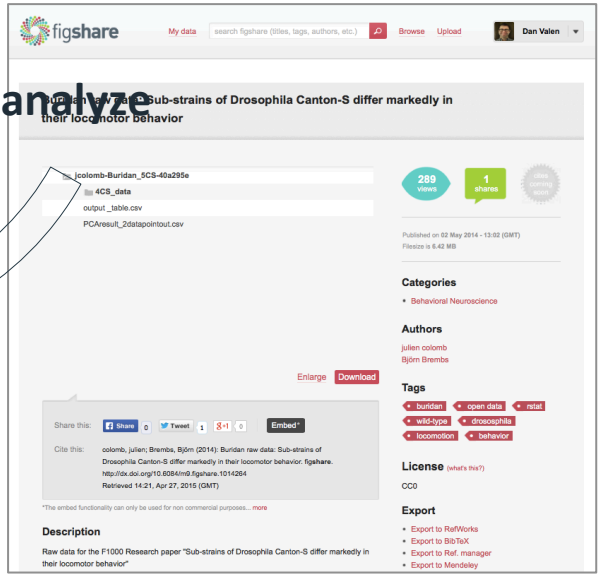
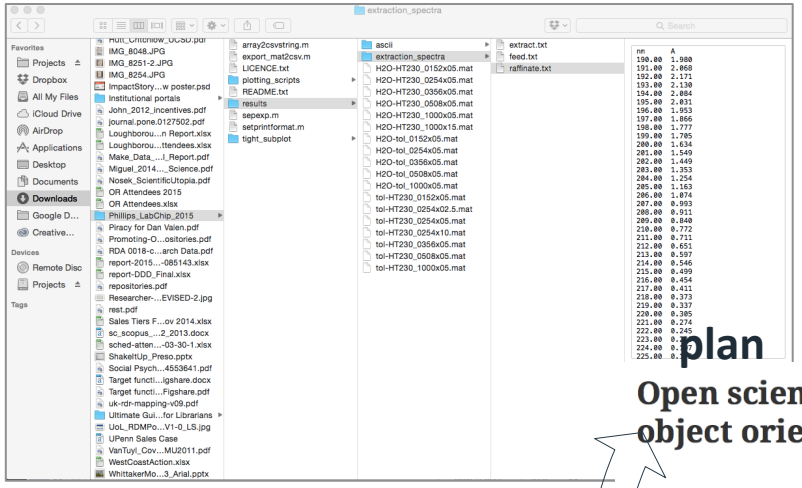


publish

Object oriented programming with figshare…



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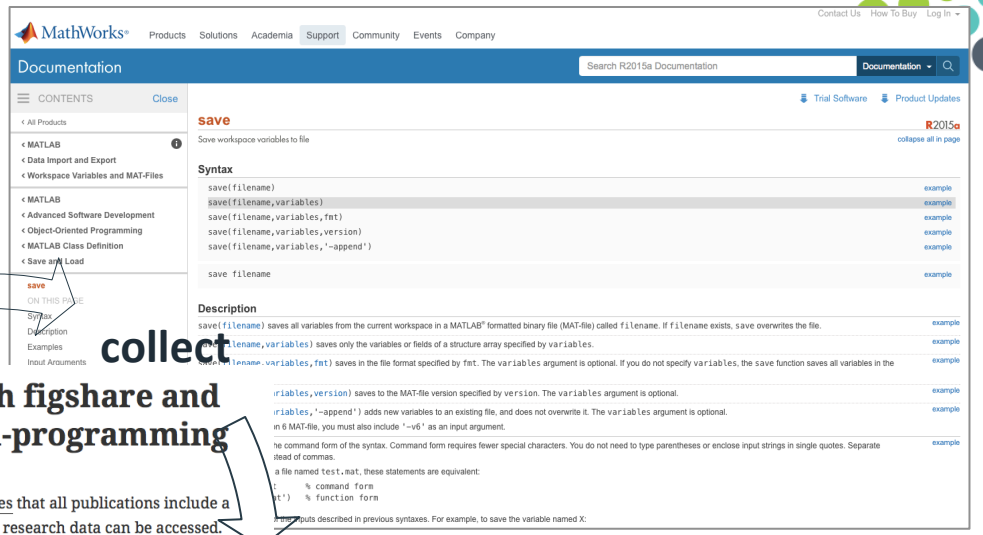
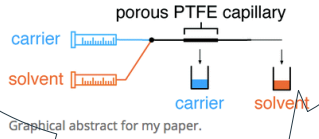


collect

Open science with figshare and object orientated-programming

From 1st May 2015, the EPSRC requires that all publications include a statement saying how the underlying research data can be accessed. Technically, you can simply include an email address to contact for the data, but I think that's hardly in the spirit of open science. In this post, I want to describe how I used object-orientated programming (OOP) and figshare to meet this requirement for my latest paper in *Lab on a Chip*. You can download the data and MATLAB code to reproduce the graphics from figshare.

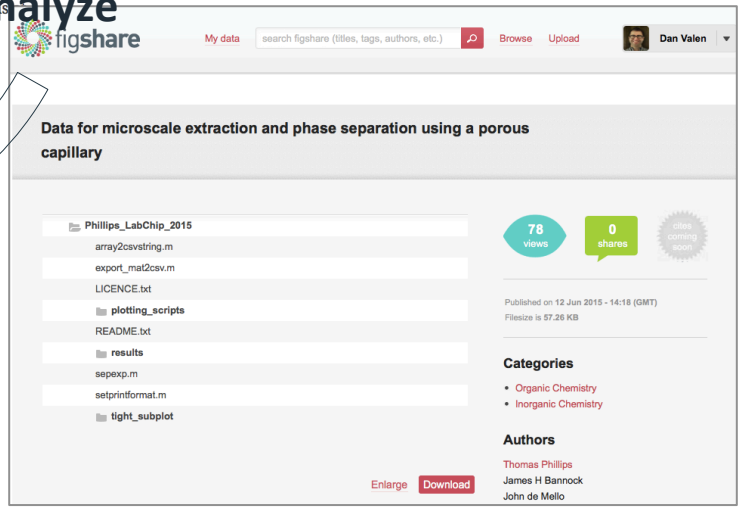
share



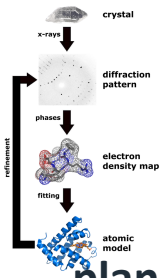
Graphical abstract for my paper.

publish

analyze

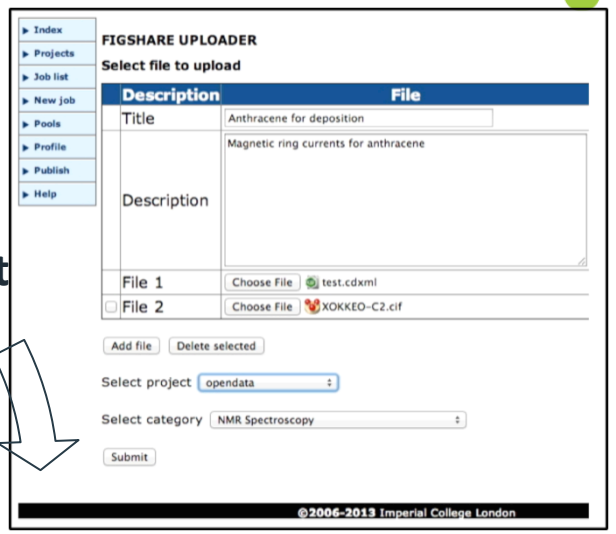


A hardware to figshare open science example



plan

collect



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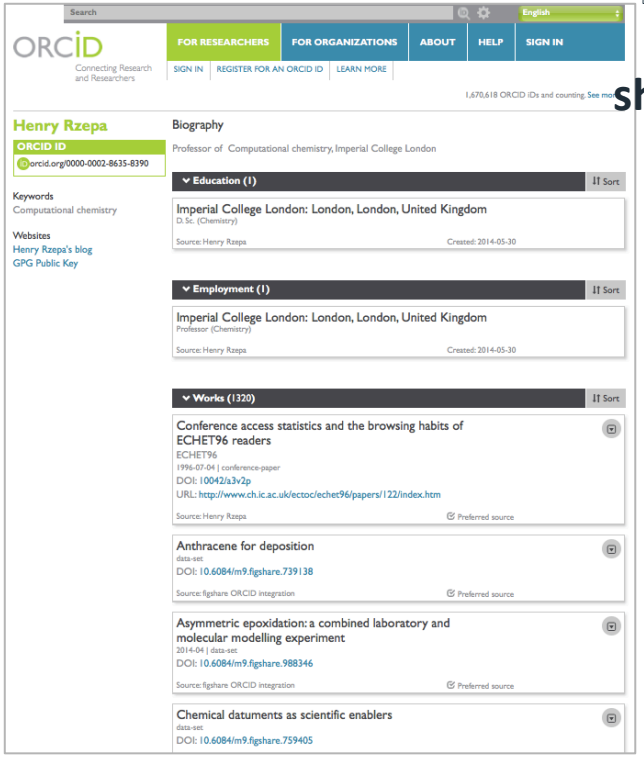
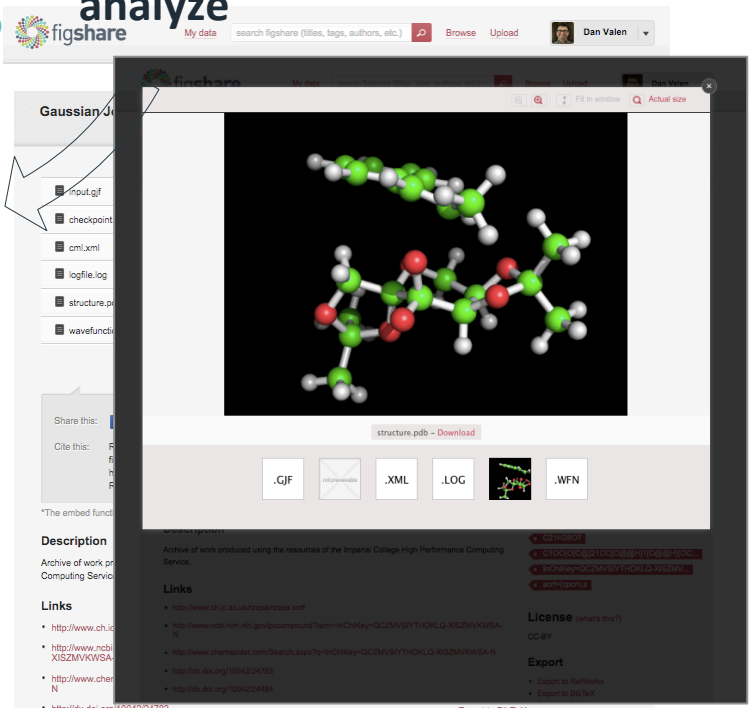


analyze

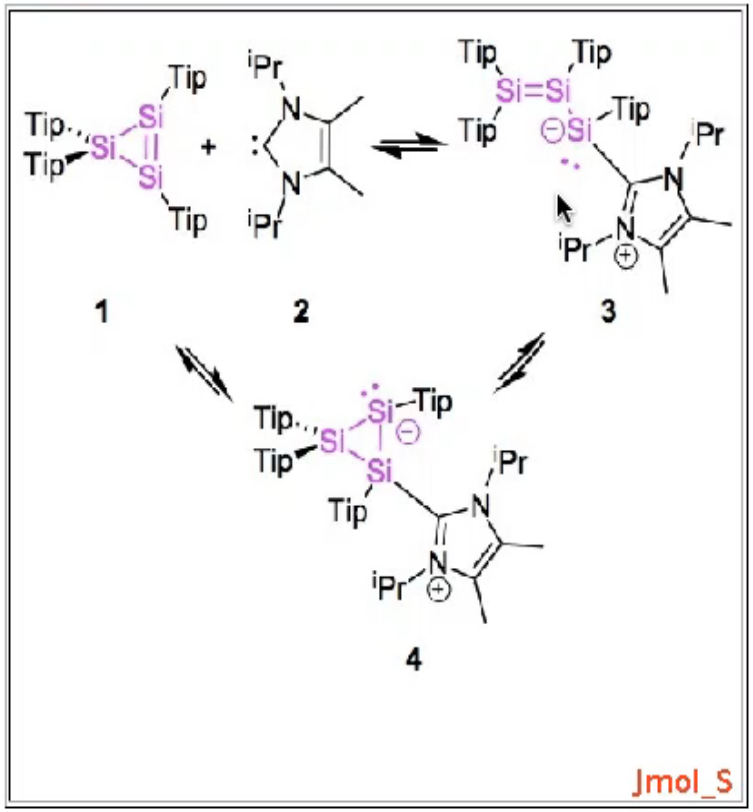


publish

Interactive data within the browser

Interactivity box. Calculated properties for Si_3 clusters.

ΔG_{298}[a]	3D model and v(Si-Si) animation	NMR and UV-Vis Spectra[e]
-3209.006417[b] -3208.558059 -3209.746651[d]	1	1 [f]
-540.412657,[b] -540.327293[c] -540.584999[d]	2	
-3749.440365,[b] -13.3[g] -3748.901271,[c] -10.0[g] -3750.297509,[d] +21.4[g]	3	3 [h]
-3749.434450,[b] -9.6[g] -3750.261410,[d] +44.1[g]	4	
-3208.984595,[b] +13.7[h]	5	



[a]Calculated using Gaussian 09, revision C.01, at the 6-311G(d,p)/6-311G(2df)[Si only] basis set. Free Energies in Hartree (relative to reactant in kcal mol^{-1}). Energies are hyperlinked to a OAI-PMH compliant Digital repository identifier, resolved as

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Volker Huch
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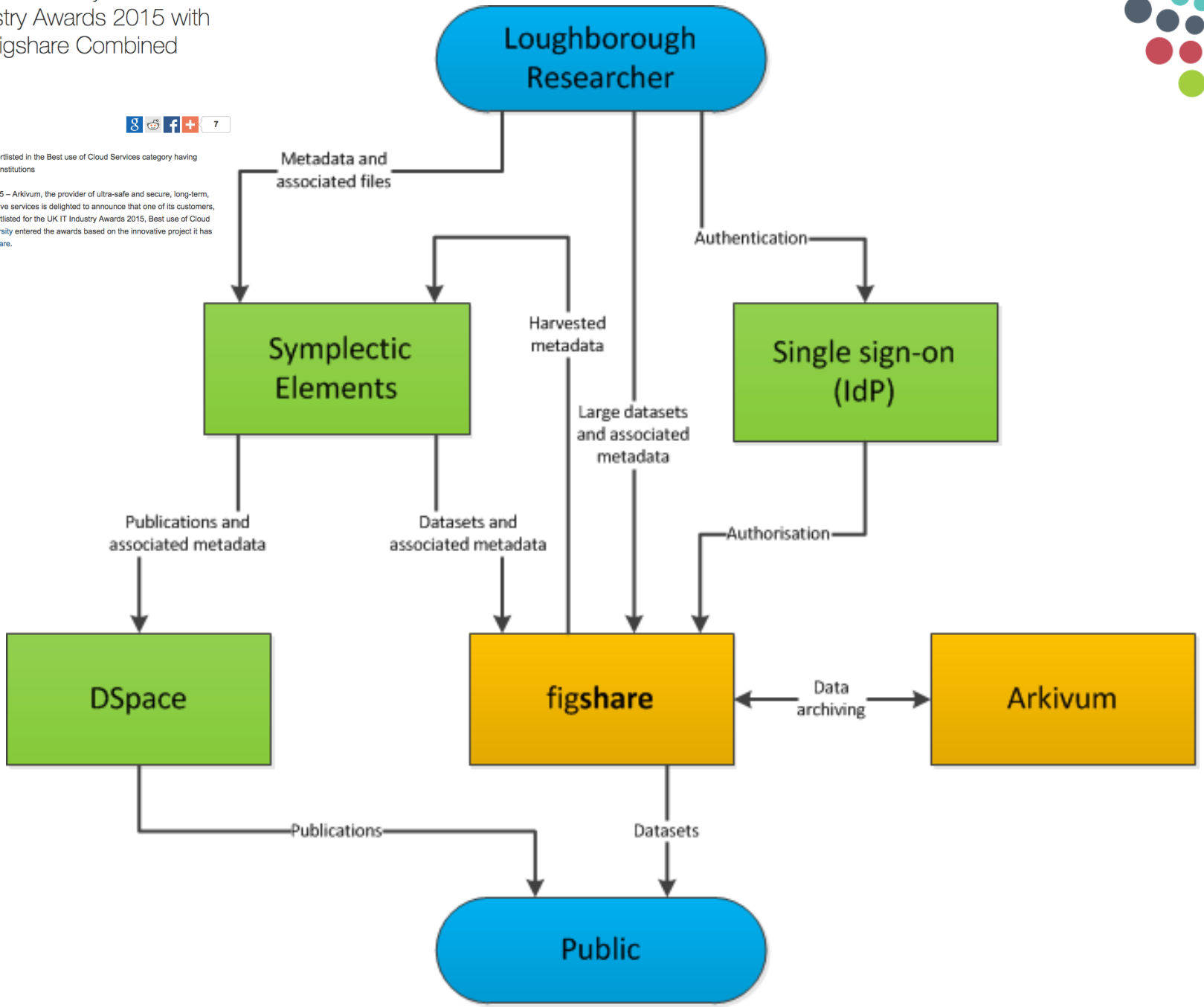
Loughborough University Shortlisted for UK IT Industry Awards 2015 with Arkivum and Figshare Combined Solution

Tuesday 25 August 2015

• Loughborough University has been shortlisted in the Best use of Cloud Services category having deployed Arkivum/100 and Figshare for institutions

Chippenham, Wiltshire, 25th August 2015 – Arkivum, the provider of ultra-safe and secure, long-term, large-scale digital data storage and archive services is delighted to announce that one of its customers, Loughborough University, has been shortlisted for the UK IT Industry Awards 2015, Best use of Cloud Services category. Loughborough University entered the awards based on the innovative project it has undertaken with both Arkivum and Figshare.



Dataverse APIs

- Data Deposit (SWORD)
- Search
- Data Access
  - Basic file access
  - Multiple files download
  - All format download for Tabular files
  - Data Variable metadata access
- Native
  - Exposes most of the UI functionality in the application via a REST-based API

Learn more about the APIs with the API Guides

Want to test out an API? Use: apitest.dataverse.org

The Dataverse Project

Current Integrations with **Dataverse** APIs

In Progress & Future Integrations with **Dataverse** APIs

Dataverse API Integrations by Users

- R package by Thomas Leeper
- Visualization of Datasets on Historical Maps by CLIO INFRA
- Anti-Slavery Petitions of Massachusetts by Garth Griffin

Interested in building off the APIs or code base?
Check out the Dataverse source code on Github.

The Dataverse Project

Support simulated and derived data

figshare

dash
DATA SHARING MADE EASY

UNIVERSITY OF CALIFORNIA GitHub

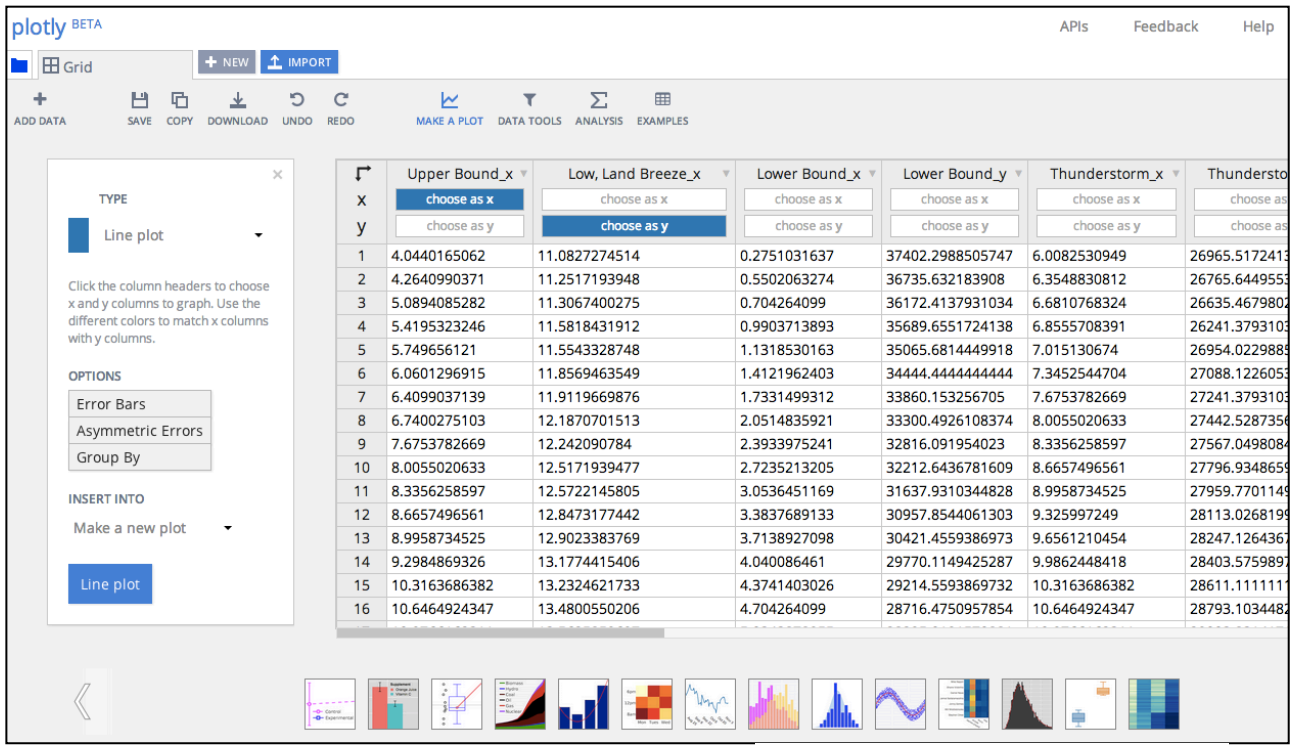
PubMed

datahub
publish, share, discover

DBpedia

Authorea

DRYAD



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UniProt

CENTER FOR OPEN SCIENCE

The Dataverse Project

MORPHOBANK
HOMOLOGY OF PHENOTYPES OVER THE WEB

DataONE
Data Observation Network for Earth

Any questions?

Get in touch!

figshare

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dan@figshare.com

The Dataverse Project

@dataverseorg

equigley@iq.harvard.edu